

ELECTIVE COURSE-7.2

URA624: IoT ENABLED AUTOMATION AND MACHINE LEARNING				
	L	T	P	Cr
	2	0	2	3.0
Course Objective: This course introduces the concepts of Industrial Internet of Things where the students are exposed to the architectures and various frameworks in IoT. The course aims to impart the knowledge to log the sensor data and to perform further data analytics. The course also introduces the fundamental theory and concepts of machine learning and its related algorithms.				
Syllabus Industrial Internet of Things (IIoT): Introduction to smart manufacturing, Internet of Things– definition and scope, Sensors for IIoT Applications- description and characteristics, various architectures of IoT, advantages and disadvantages. Real life examples in manufacturing industries, challenges and benefits in implementing Industrial IoT. Digital twin principal and their application in smart manufacturing. IIOT Protocols: Industrial IoT protocols and communication methods. Computing Methods: Basic concepts of programming language: values, types, variables, operands and expressions, conditional statements, loops, functions. Machine Learning: Introduction, types of machine learning - supervised learning and unsupervised learning, classification and regression. Classification Algorithms: Logistic regression, support vector machines, decision trees, random forests. Regression Algorithms: Linear regression, Lasso and Ridge regression, Support vector regression, Decision trees regression. Machine Intelligence: Machine Vision, Artificial neural network with applications for mechanical sciences. Case studies on industrial automation: Practical applications of IIoT and machine learning in manufacturing industries.				
Laboratory Work (if applicable) Google Colaboratory, Presentations and research assignments in: Jupyter notebook – preferred by mechanical industry, Computing: Python (Open source) – preferred by mechanical industry.				
Course Learning Outcomes (CLOs) The students will be able to:				

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1. analysis data of IoT sensors of industrial mechanical systems.
2. understand the basic concepts of machine learning algorithms and identify machine learning techniques suitable for the given problem.
3. design an approach to leverage data using the steps in the machine learning process.

Text Books

1. Minoli, D., *Building the Internet of Things with IPv6 and MIPv6: The Evolving World of M2M Communications*, Wiley (2013).
2. Trevor Hastie Robert Tibshirani Jerome Friedman, “*The Elements of Statistical Learning, Data Mining, Inference, and Prediction*”, (2nd Edn.), Springer, 2014.
3. Sonka M., Hlavac V. and Boyle R., *Image Processing, Analysis and Machine Vision*, Thomson Learning, (1993) 1st ed.
4. Bishop M., C., *Pattern Recognition and Machine Learning*, Springer-Verlag (2011) 2nd Edition.

Reference Books

1. Russel S., Norvig P., *Artificial Intelligence: A Modern Approach*, Prentice Hall (2014) 3rd ed.
2. G James, D. Witten, T Hastie, and R. Tibshirani, “*An Introduction to Statistical Learning: with Applications in R*”, Springer, 2013.
3. Russel S., Norvig P., *Artificial Intelligence: A Modern Approach*, Prentice Hall (2014) 3rd ed.
4. G James, D. Witten, T Hastie, and R. Tibshirani, “*An Introduction to Statistical Learning: with Applications in R*”, Springer, 2013.

Evaluation Scheme:

S.No.	Evaluation Elements	Weightage (%)
1.	MST	30
2.	EST	45
3.	Sessional (Lab/Assignments/Projects/Tutorials/Quizes)	25

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